

Evolving Robot Morphology with Manipulation Learning Inspired by Human Hand Evolution

Bangchu Yang, Li Jiang*, Wenhao Wu, Ming Cheng, *Member, IEEE*, Kening Gong

Abstract—The evolution of human hands is intricately linked to their morphologies, behaviors, and selection pressures. This biological evolution process provides critical insights for robot design, facilitating robot evolution through environmental interaction and skill learning. In this context, we propose a Baldwin-effect-inspired framework that endows robots with the capacity for long-term enhancement in response to environmental changes. This framework integrates large-scale population evolution and morphology-aware manipulation learning. Each robotic hand undergoes extensive testing of its grasp and manipulation abilities with various objects. Leveraging this framework, these hand agents validate several prior hypotheses related to morphological embodied intelligence: the relative advantage of human-like segment models in manipulative potential, the impact of the environment on the final evolved morphologies, and the influence of morphology difference on the agent's learning ability. These phenomena are not independent, as multiple agents with different segment models exhibit convergent evolution and morphological specialization. Furthermore, this natural evolutionary capability has been implemented in three-, four-, and five-fingered reconfigurable robotic hands, each following unique evolutionary trajectories that evolve towards higher bilateral symmetry. The evolved prototypes have demonstrated superior performance, surpassing anthropomorphic designs by human experts.

Index Terms—Baldwin effect, manipulation learning, population evolution, robot evolution, reconfigurable robotic hands.

I. INTRODUCTION

Throughout millions of years of human evolutionary history, there has been a profound and intricate interplay among hand evolution, tool usage, and brain development [1, 2]. These have facilitated the transition of humans from the Paleolithic era to modern civilization, evolving hand morphologies and manipulative functions [3, 4]. This evolutionary process is reminiscent of a Baldwin effect, wherein learned behaviors can influence the evolutionary trajectory via selection pressures over generations [5].

Rather than undergoing evolution as observed in biological organisms, robotic hands are primarily designed through biomimetic approaches that replicate human hand

morphology, gestures, and grasping functions. In terms of morphology, many state-of-the-art human-like robotic hands [6-17] replicate key anatomical features of the human hand, such as the proportions and lengths of the phalanges [9], as well as the combination of rigid bones and soft tissues [10]. For posture replication, human hand postural synergies are extracted using techniques such as principal component analysis. These synergies can be represented through appropriate joint coupling ratios, enabling simplified control of underactuated robotic hands [6-8, 11, 12]. This approach allows for the reproduction of the majority of human hand postures using fewer actuators. For grasping functions, robotic hands are often optimized using task-specific evaluations. Mechanical metrics such as force closure and form closure are widely used to assess and enhance grasp stability, thereby guiding the design of finger geometry [18] and tendon routing [19] in underactuated hands. These methods perform well in grasping scenarios or predefined postures but do not extend to more complex tasks such as in-hand manipulation learning.

An emerging paradigm in robot design focuses on co-optimizing morphology and control, drawing inspiration from biological evolution. This method aims to explore the mutual adaptation between morphology and control strategies. Representative works include voxel-based soft robots in two-dimensional spaces [20], rod-based bipedal and multipedal agents [21-24], and virtual agents with diverse topological configurations [25, 26].

These co-optimization studies commonly adopt a two-level nested optimization architecture, where an outer loop evolves the agent's morphology and an inner loop optimizes its control policy. The outer loop primarily employs evolutionary algorithms, such as quality-diversity (QD) methods [27], covariance matrix adaptation evolution strategy (CMA-ES) [28] or tournament-based steady-state evolution [25]. Some approaches further incorporate gradient-based optimization to improve efficiency [21]. In the inner loop, increasing attention has been given to training universal controllers that can generalize across multiple morphologies. Knowledge distillation techniques have been proposed to integrate multiple morphology-specific controllers into a single policy capable of zero-shot generalization to unseen morphologies [27]. Modular controllers have also been proposed to enhance transferability across different embodiments [29, 30], such as modular robots [31]. In addition, transformers [26] and graph grammars [24] have been used for encoding morphology design and joint motion.

Building on these pioneering studies, some work explores applying co-optimization to robotic hardware design [32, 33].

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For instance, some works have modeled grippers with simplified cubic-form fingers to investigate the influence of finger position, orientation, and length on grasping performance [34]. Alternatively, fingertip geometries have been optimized using end-to-end differentiable frameworks, with the resulting components fabricated via 3D printing. However, these designs typically lack actuators and sensors, limiting their functionality [35]. In contrast, the Hardware as Policy (HWaP) framework [33] presents a more comprehensive scheme that optimizes joint stiffness in an underactuated robotic hand, with corresponding real-world implementations. Nonetheless, the authors acknowledge that the system remains relatively simplistic, being limited to a single grasping task and employing only one actuator. This scheme can also be applied to locomotion robots, where robot bodies and controllers are evolved in simulation and the best designs are then 3D-printed to validate their locomotion in the real world [36].

In this work, our goal is to explore whether biologically inspired co-optimization methods remain effective when applied to high-degree-of-freedom robotic hands and complex manipulation tasks, and to further bridge the gap between co-optimization theory and real-world robotic hands. To this end, we propose a biologically inspired framework that integrates hand morphology evolution with manipulation learning. Using this framework, we trained 20-degree-of-freedom hand agents across 16 grasping scenarios and 8 manipulation scenarios. For parallel execution, the 16 grasping scenarios were instantiated twice to form 32 parallel grasping environments, and the 8 manipulation scenarios were instantiated four times to form 32 parallel manipulation environments per morphology. Thus, each hand morphology was shared across 64 parallel environment instances. These scenarios define Task Set 1, which is used by default unless otherwise stated; details are provided in Section III.A. The evolutionary process exhibited instances of convergent evolution and morphological specialization. On the hardware, we developed a reconfigurable robotic hand prototype featuring variable link lengths and a palm structure capable of supporting a flexible number of fingers. The evolved morphologies were further transferred to the real-world reconfigurable robotic hand platform.

Our contributions are summarized as follows:

1. **Bioinspired framework:** we extend nested co-optimization to in-hand manipulation scenarios using high-degree-of-freedom hands with reconfigurable morphologies.
2. **Morphological validation:** this framework validates hypotheses proposed in prior studies regarding the task-specific optimal morphologies and further reveals the interdependence among morphology, learning, and the environment.
3. **Hardware transferability:** we develop a fully actuated, reconfigurable robotic hand prototype with a variable number of fingers and link lengths, enabling transfer of the optimized morphologies to the hardware platform.

II. METHODS

A. Framework Overview

In evolutionary biology, the Baldwin effect describes how learned behaviors can influence evolution through selection pressures [5]. In complex environments, traits that facilitate skill acquisition are favored by natural selection over generations, gradually leading to the development of learning-supportive morphologies within the population [5, 25, 37, 38]. In human evolution, precision grip and tool use are believed to have contributed to anatomical changes in the hand [39]. One notable example is the evolution of a longer and more opposable thumb [4, 40]. This enhanced opposability significantly improved the precision grip, a capability that is especially refined in humans compared to other primates. In addition, the demand for manipulation drives the differentiation of hand muscles and tendons [41], a phenomenon often referred to as manipulative specialization, which in turn supports the use of complex tools and skilled object manipulation.

Inspired by this biological process, our work introduces a co-optimization framework that simulates a Baldwin-like mechanism in robotic hands by integrating environmental setup, morphological adaptation, manipulation learning, and task-driven fitness evaluation within the evolutionary process (Fig. 1). Therefore, tens of thousands of parallel environments were created to support the evolution and learning of a large population of robotic hands. Unlike organisms possessing individual cognition, the centralized morphology-aware controller is shared among the large-scale hand population. This controller is trained via the Transformer-based Proximal Policy Optimization (PPO) reinforcement learning model, and makes decisions according to the feedback of morphological and motion information. According to environmental fitness, the population is divided into survivors and non-survivors. Among the survivors, the fittest individuals, referred to as parents, undergo further mutation and reproduction. The number of new individuals matches the count of non-survivors, thereby maintaining a constant population size dynamically. Besides the large-scale population, each hand undergoes testing over long durations. The count of rollouts ranges from tens to thousands, influenced by differences in morphology, skill learning, and environmental factors. During each rollout, grasping quality and manipulative precision are calculated to update the common adaptive features across grasping and manipulation tasks. These enhancements facilitate the evolution of the hands into new stable morphologies and a more adaptive population (Fig. 2).

The framework follows the general paradigm of a two-level nested co-optimization architecture, combining morphology evolution with transformer-based policy learning. In the two-level nested co-optimization framework, each generation has two strictly separated phases. In the inner-loop phase, for every candidate morphology in the current population, the shared Transformer-based controller is trained using reinforcement learning to update the Transformer parameters, without performing any evolutionary selection or mutation.

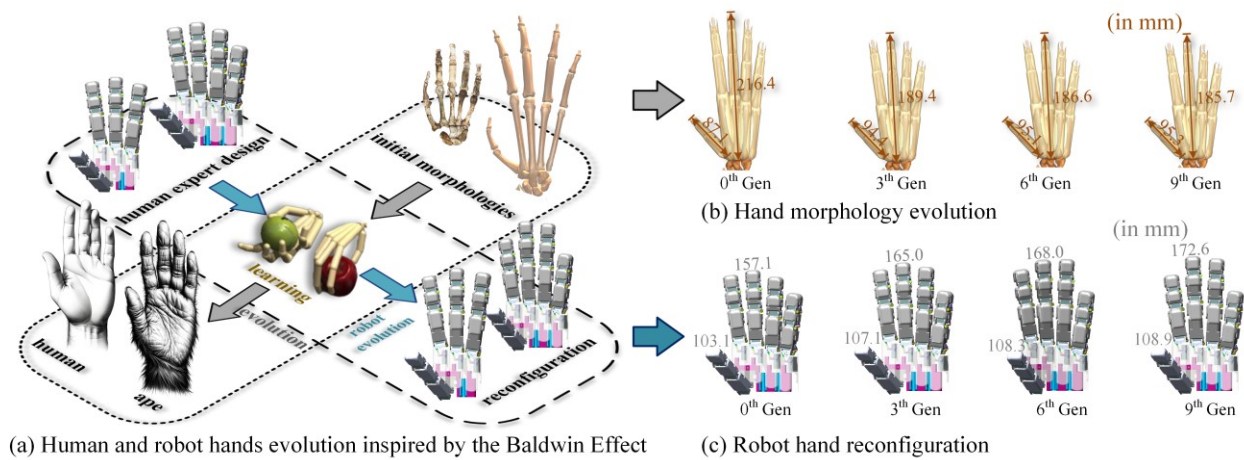


Fig. 1 Overview of hand evolution in humans and robots inspired by the Baldwin effect. (a) A conceptual illustration of the Baldwin effect: learned behaviors, such as grasping or manipulation, influence morphological evolution by selection pressures. This process, observed in primate evolution, inspires the co-optimization of morphology and learning in robotic hands. (b) Virtual hand morphologies evolve over generations, with adaptations that enhance in-hand manipulation and grasping. (c) The evolutionary method is employed for robotic design, and transfer the optimized morphologies from simulation to a reconfigurable hardware platform, enabling the physical realization of bioinspired morphological evolution.

Then, in the outer-loop evaluation phase, these Transformer weights are frozen and used to compute morphology-level fitness (grasp quality and number of successful reorientations) via rollouts, so individual evaluations are not influenced by policy updates or by their evaluation order within a generation. A related approach is METAMORPH [26], which also employs a transformer-based controller but is designed for legged robots. In contrast, our controller targets in-hand manipulation using high-degree-of-freedom robotic hands, resulting in significant differences, such as custom morphology-motion embeddings, spatial positional encoding, and GPU acceleration, which enable efficient large-scale simulation of high-degree-of-freedom hand dynamics on a single GPU. To enable a direct comparison with prior work, we adapted METAMORPH to our manipulation environments and tested the effect of positional encoding. We also analyzed the scalability advantages of GPU-based simulation. These evaluations are presented in Section II.G.

B. Morphology-Aware Controller

The framework is implemented by GPU-based parallel population simulation using Isaac Gym [42]. The morphological, action, object, and reward information collected by parallel local environments are aggregated into two unified buffers: the grasping and manipulation buffers, for training shared models. This shared model can be deployed in each local environment during the decision-making process. GPU-based population simulation can provide large-scale real-time data for PPO [43] training. The actor and critic networks of PPO share features extracted by a Transformer. The positional encoding in the Transformer uses spatial information from the hand structure, for example, the distal phalanx of the thumb is encoded as (1, 3), due to the richer

information than absolute or learnable positional encoding. During decision-making, the above information undergoes normalization, linear projection, multi-head attention, etc. The output of the Transformer is flattened and fused with external information, which primarily includes the spatial position and velocity of the object and fingertips, the object's orientation, size, and shape primitives. The position, velocity, and control feedforward of the wrist joint are also input as external information. Drawing upon the foundational work introduced in Reference [26], this framework integrates and refines the morphology-aware controller initially proposed for multipedal robot evolution. Furthermore, we conduct a comprehensive evaluation and comparison of the original controller with our version within the context of our large-scale population and evolution task in Section II.G.

Before controller training, the controller reloads the model from the last generation to accelerate learning and reduce training time. During training, agents interact with their respective environments in the same manner as during decision-making. To further enhance policy robustness, we employ domain randomization: the mass and friction, joint stiffness and damping, and external force perturbations applied to the objects are randomly varied. The controller is trained using Proximal Policy Optimization (PPO). During PPO-based training, the grasping and manipulation controllers are trained independently, although they share the same morphological parameters, because they use different reward functions: the grasping controller's reward is primarily determined by the fingertip-to-object distance, the height to which the object is raised, and whether it reaches the target position, whereas the manipulation controller's reward focuses on object orientation and position. Both controllers also impose penalties for excessive actuator torque.

Note that the fitness evaluation differs from the reward functions. In the evolutionary outer loop, hand morphologies are ranked only by the fitness function, not by the reinforcement learning rewards. Specifically, the grasping fitness (grasp quality) is computed from the final contact configuration using a classical grasp wrench space metric—the smallest singular value of the grasp matrix G —which is widely used as an indicator of grasp robustness and isotropy, as it reflects how uniformly the contact wrenches can resist external disturbances. This metric belongs to the family of wrench space approaches that evaluate grasp stability based on contact locations and contact models. The grasp-stability score is set to 0 if the object is dropped. This morphology-level fitness is evaluated at the end of the rollout (evaluation phase) and does not depend on terms such as lifting height or target reaching. The manipulation fitness is the count of successful reorientations. Reorientation is a classical in-hand manipulation task, in which the object is rotated to a new, target pose while remaining stably grasped. By specifying a

sequence of intermediate reorientation targets, the hand can perform continuous and dexterous in-hand manipulation of the object, such as pen spinning or rotating a cube.

By contrast, the reward functions are used solely inside the PPO-based inner loop to facilitate policy learning. Some reward terms are assigned relatively small weights as dense shaping signals to mitigate reward sparsity and to guide the controller toward discovering stable grasp-and-lift or manipulation behaviors. These reward terms help the policy learn to grasp or manipulate objects, but they are not the criteria by which hand morphologies are selected in evolution. After each rollout, the evolutionary algorithm evaluates each morphology using the grasp-matrix-based fitness (for grasping) or the reorientation-count-based fitness (for manipulation), which together provide a more direct measure of hand design quality.

In some borderline cases, the object can still slip or be dropped during lifting or motion, especially in early training. However, such failures are rare in the rollout phase used for

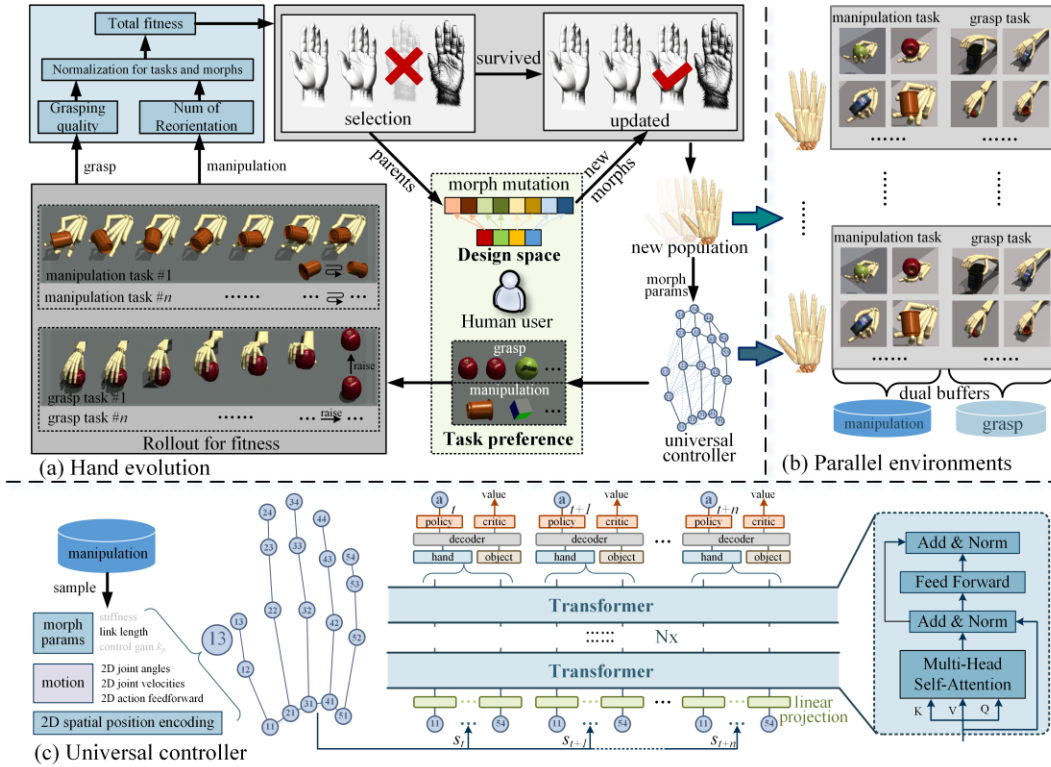


Fig. 2 Two-level nested co-optimization framework. (a) The outer evolutionary loop implements a generational evolutionary algorithm with elitism and mutation-only variation. In each generation, a population of N morphologies is evaluated using task rollouts, and fitness is computed from grasp quality and manipulation performance, normalized across tasks and morphologies. The top- k morphologies are retained as elites and copied to the next generation. A subset of these elites is further selected as parents, and the remaining individuals in the population are created by mutating these parent morphologies. The “Human user” in the task preference module indicates user-specified task preferences, which determine the types and numbers of grasping and manipulation objects, as well as which morphology parameters are subject to mutation and the corresponding mutation ranges. These factors jointly influence the final hand morphology and are further illustrated in Fig. 3. (b) Together with (c), this forms the inner control loop. In each generation, newly generated morphologies from (a) and the centralized controller from (c) are deployed into parallelized simulation environments to perform grasping and manipulation tasks. The resulting interaction data are stored in dual replay buffers. (c) The centralized Transformer-based universal controller, trained with PPO. Morphology and motion data stored in (b)’s buffers are encoded into graph representations and mapped to transformer input tokens. These tokens are used to train the policy and value networks for both manipulation and grasping tasks.

fitness evaluation.

Detailed methodologies for computing the reward and fitness functions are provided in Section II.D.

C. Task Preferences

To manage these environments, the framework defines "task preferences" to determine the in-hand manipulation or grasping task proportions, the scope of object sets, and the frequency of object appearances. These can be acquired from the frequency of manipulation [44] and grasping [45] in daily life. Task preference is one of the determinants in shaping the final hand morphology. Like human evolution, minor preferences tend to promote the broad applicability of morphology, while significant preferences lead to morphological specialization. Unlike human evolution, these specialized robot morphologies can be further adjusted according to task changes in the field of reconfigurable hands. In terms of morphological variation, potential variations in morphological parameters, such as segment length, can be explored through design space. However, the development of real-world robots [46, 47], necessitates trade-offs among various factors, including but not limited to performance, practicality, aesthetics (particularly for prosthetic hands), and expandability (modularity). Thus, it is necessary to employ the mapping relationship between design space and morphological parameters to determine the scope and proportion of adjustments. This ensures that the evolution of prototypes is within the constraints of design conditions, such as the modular design of fingers (Fig. 3). The preferences for specific tasks by humans, along with the available design space, serve as critical factors in determining the ultimate morphology of the hand.

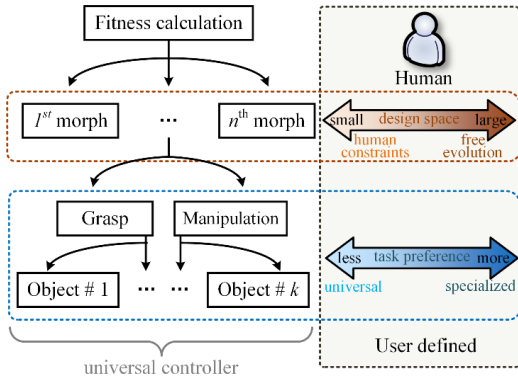


Fig. 3 Artificial Constraints. Design space and task preference play a pivotal role in determining the final morphology of the hand evolution.

D. Fitness and Optimization

Fitness is determined by grasping and manipulation tasks with multiple objects. For in-hand manipulation tasks, the object is on the palm, and its dexterity is demonstrated through coordinated finger gaits; for grasping tasks, the object is below the palm, and it can maintain grasp stability during the arm's

movement.

The calculation of grasping quality uses a classical model [48, 49] to assess stability. An object is considered successfully grasped when it is lifted from the ground to a target height. We use the smallest singular value of the grasping matrix G , the count of successful grasps m , and the quantity of objects n to compute the grasping fitness function f_G .

$$f_G = \sum_{j=1}^n \text{norm}(\sum_{k=0}^m \min(\det(\sqrt{G_{jk}G_{jk}^T}))). \quad (1)$$

$$G = \begin{bmatrix} I_{3 \times 3} & I_{3 \times 3} & \dots & I_{3 \times 3} \\ [x_1]_{\times} & [x_2]_{\times} & \dots & [x_n]_{\times} \end{bmatrix}. \quad (2)$$

$$[x_i]_{\times} = \begin{bmatrix} 0 & -x_i^{(3)} & x_i^{(2)} \\ x_i^{(3)} & 0 & -x_i^{(1)} \\ -x_i^{(2)} & x_i^{(1)} & 0 \end{bmatrix}. \quad (3)$$

Where x_i is the coordinate of the i th contact point, the function norm normalizes the grasping quality of objects to the $[0, 1]$ interval, ensuring that different objects contribute equally to the grasp fitness. Unlike the grasp fitness, the reward function R_G also needs to consider the immediate rewards, as discussed in the reference [42, 50].

$$R_G = \lambda_1 R_{tip} + \lambda_2 R_{lift} - \lambda_3 P_{act_G} + R_{bonus_G}. \quad (4)$$

$$R_{tip} = \max(0, \min(n(0.5, -\Delta \sum d_{tip}))). \quad (5)$$

Where R_{tip} is the reward for the distance between the fingertip and the object, $\sum d_{tip}$ is the sum of distances between all fingertips and the object, $\Delta \sum d_{tip}$ represents the difference between this cumulative result and the previous one, and the final result is limited to the $[0, 0.5]$ interval. R_{lift} is the reward for the height to which the object is lifted, P_{act_G} is the penalty for excessive actions, with its coefficient adapted from [50], and upon reaching the target height, R_{bonus_G} is awarded, valued at 300.

For in-hand manipulation tasks, once the object reaches the target orientation, a new target orientation is generated, resulting in the fingers performing continuous manipulation gaits to track the continuously changing target orientations. The object is reset if it falls or fails to reach the target orientation for a long duration. Based on the above process, the manipulation fitness function f_M should reflect the success rate of reorientation and the continuity of manipulation.

$$f_M = \sum_{j=1}^n \text{norm}(m_{rot}). \quad (6)$$

Where m_{rot} represents the count of successful reorientations, the manipulation reward function refers to OpenAI [42, 51].

$$R_M = \lambda_4 R_{pos} + \lambda_5 R_{rot} - \lambda_6 P_{act_M} + R_{bonus_M}. \quad (7)$$

Where R_{pos} is the distance from the target position, R_{rot} is the difference from the target orientation, P_{act_M} is the penalty for excessive actions, with its coefficient adapted from [51], and

R_{bonus_M} is awarded once the object reaches the target orientation.

Fitness values are normalized along the task dimension, ensuring that each task contributes equally to fitness. The total fitness is further normalized along the morphological dimension. For the hand morphology δ , the fitness $F(\delta)$ is

$$F(\delta) = \text{norm}(f_c) + \text{norm}(f_M). \quad (8)$$

Hand morphologies are selected based on their fitness $F(\delta)$ rankings, with the top-performing designs chosen for reproduction in subsequent generations. This selection mechanism promotes the propagation and refinement of successful morphologies over time. However, computational resource constraints, particularly memory limitations, impose an upper bound on the number of concurrent tasks. For example, an NVIDIA RTX 4080 GPU can process approximately 32,768 tasks simultaneously. To maintain a consistent population size while focusing on promising candidates, lower-fitness morphologies are eliminated. The emerging morphologies in the next generation m_{new} are obtained by inheriting the elite morphologies from the current generation m_{old} and adding uniform noise $U(-\delta/2, \delta/2)$ in the range:

$$m_{new} = m_{old} + U(-\delta/2, \delta/2). \quad (9)$$

Here, δ controls the exploration strength through the perturbation magnitude. We do not introduce a separate mutation rate to preserve parent morphologies, since the top- k fittest individuals are already copied unchanged to the next generation, as detailed in the caption of Fig. 2.

E. Morphological Adjustment and Representation

For morphological construction, we used a 50th-percentile male hand [52] as the base model for the virtual human hand model. In contrast, the virtual reconfigurable robotic hand is

constructed to match the real-world hardware [53], which supports adjustable mechanical parameters: joint stiffnesses, palm curvatures, finger placements, and phalanx lengths. In this study, however, the evolutionary search is restricted to morphology, and we optimize only the 19 morphology-related parameters listed in Table I, while the remaining parameters are kept fixed. Both hand models are built using MJCF, a robot description format developed for MuJoCo and compatible with Isaac Gym.

TABLE I
THE MECHANICAL PARAMETERS AND RANGES

	F1	F2	F3	F4	F5
Finger Location	0~10	0~10	0~10	0~10	0~10
Proximal Phalanx	0~25	0~25	0~25	0~25	0~25
Middle Phalanx	0~15	0~15	0~15	0~15	0~15
Distal Phalanx	0~5	0~5	0~5	0~5	0~5

F1 to F5 denote the range from the thumb to the little finger; Finger locations and proximal/middle/distal phalanx lengths are in millimeters (mm); All parameters are continuous.

For morphological adjustments, phalanx lengths and finger placements are primarily modified by adjusting the "body" and "geom" elements in the MJCF file during the evolutionary process. The MJCF file can be parsed and modified using Python's ElementTree package. The updated MJCF file is subsequently reloaded into the newly created environments. For morphological representation, we construct a graph based on the hand's spatial configuration. The morphological information stored in the npy files is mapped to corresponding nodes in the graph, along with joint motion data and spatial position encodings, as illustrated in Fig. 2c. For instance, node (1, 3) denotes the first finger (thumb) and its third phalanx (distal phalanx). This node records phalanx length as well as the angle, velocity, and feedforward action of the interphalangeal (IP) joint.

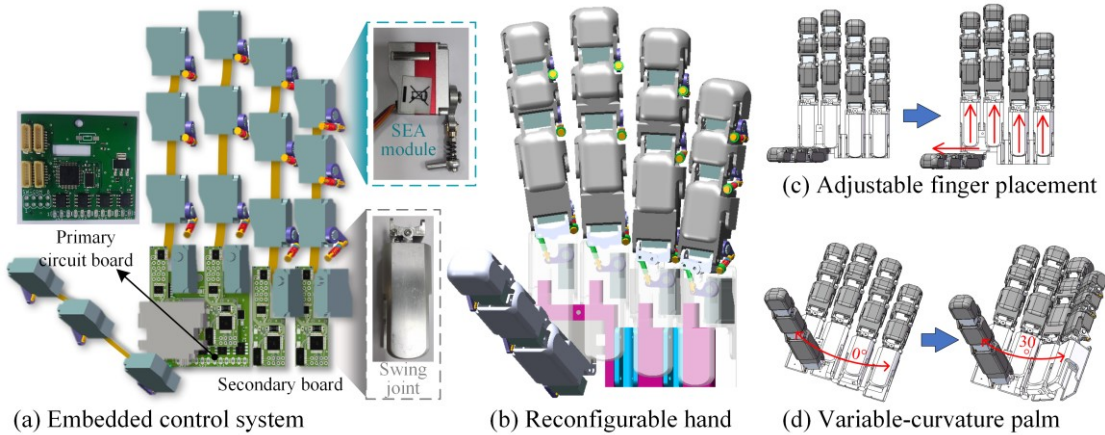


Fig. 4 The Robotic Reconfigurable Hand Platform. (a) presents the embedded control system, where the primary circuit board communicates with the metacarpal (MC) boards via SPI. The hand is fully actuated by 20 motors, with each flexion joint incorporating a Series Elastic Actuator (SEA). (b) showcases the external appearance of the reconfigurable hand. (c) the adjustable mounting positions of the five fingers are highlighted, with red arrows indicating the directions of adjustment. And the phalangeal length can also be adjusted using the same mechanism. (d) demonstrates how the palm curvature can be modified using pairs of fixable rotation components.

F. Robotic Reconfigurable Hand Platform

The framework is deployed on hand prototypes with three, four, and five fully actuated fingers, facilitating the evolution of all the finger lengths. These fingers are composed of reconfigurable modules characterized by adjustable segment lengths, and are equipped with Series Elastic Actuators (SEAs), motor-current sensors for torque estimation, and joint-position sensors. Furthermore, the design of the palm supports a variable number of fingers.

The hand's configuration can be modified to three-, four-, or five-fingered variants using fixable carpometacarpal (CMC) joints in the palm. Phalangeal length and finger attachment positions are adjusted via fixable sliding components. This robotic reconfigurable hand was first presented in a previous work [53], and we have further updated this robot to explore morphological evolution in this study.

This fully actuated hand employs 20 motors, with each flexion joint utilizing a Series Elastic Actuator (SEA) - a four-bar linkage mechanism with a spring. The hand's embodied system follows a primary-secondary architecture, comprising one primary controller and four secondary controllers. The primary board communicates with the metacarpal (MC) boards via SPI (Serial Peripheral Interface), while each MC board controls four motors and estimates joint torque through current readings (Fig. 4).

G. Framework Parameters and Comparisons

We examined the impact of various hyperparameters, such as population size (Fig. 5) and controller (Fig. 6), on the framework's performance, and investigated their effects across manipulation and grasping scenarios. Regarding population scale, we observed a pronounced emergence of manipulation ability upon reaching a critical scale; however, this phenomenon was not as evident in grasping learning.

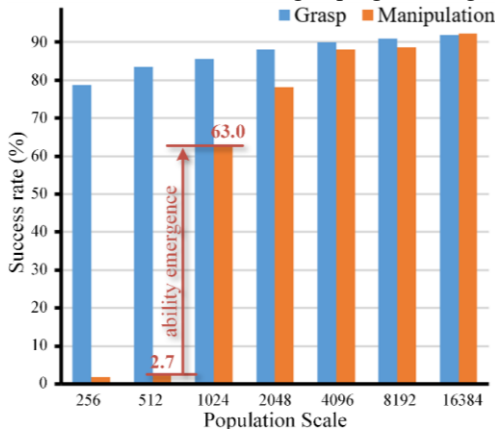


Fig. 5 Population size and ability emergence. When the population size reached a certain value (between 512 and 1024), a marked emergence of manipulation ability was observed. However, this phenomenon was not significantly manifested in grasping.

The success rate used in Fig. 5 is defined as follows: a grasping rollout is considered successful when the target

position is reached, and a manipulation rollout is considered successful when the target orientation is reached. In each generation, for each morphology in each parallel environment, we perform hundreds of grasping or manipulation rollouts. For each morphology, the success rate is first calculated in each environment as the fraction of successful attempts out of the total number of attempts, and is then averaged across its 64 parallel environments.

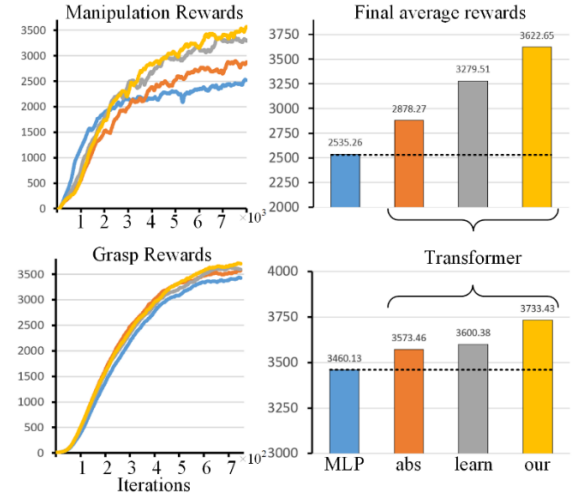


Fig. 6 Hand performance across three positional encodings.

From left to right, there are the multilayer perceptron, the Transformer with absolute positional encoding, learnable positional encoding, and spatial positional encoding that we used. The left panels show PPO training curves for grasping and manipulation under a fixed morphology. The right panels show the final average rewards, computed as the mean of the last 100 points of the corresponding learning curves.

For controller design, Fig. 6 compares an MLP baseline with three Transformer-based controllers, all evaluated on the same 32 grasping and 32 manipulation tasks under the same fixed morphologies. This comparison only concerns the inner-loop controllers and does not involve the outer-loop evolution. The METAMORPH controller [26] uses two types of positional encodings (absolute and learnable) and was originally proposed for legged robots; here we adapt it to our grasping and manipulation tasks for comparison. Our controller instead employs a hand-centric spatial positional encoding that embeds the hand morphology, including the finger index (chain) and the depth in the kinematic tree. The learning curves for grasping and manipulation, together with the final average rewards, show that all Transformer variants outperform the MLP baseline and that our spatial encoding achieves the highest final rewards on both tasks, with an even larger advantage in the more challenging manipulation scenario.

III. RESULTS

A. Morphological Change under Selection Pressure

Previous studies have assessed the manipulative potential based on the segment lengths of fossilized hand specimens [40, 54] and inferred the factors influencing morphological

specialization [55]. The framework can serve as a validation tool for these inferences. Leveraging this tool, we can clarify the impact of the selection pressures on the transition from initial to inferred specific morphologies, as well as the numerous evolutionary details: (1) The interplay between diverse factors and morphological evolution, with a particular focus on selection pressures (task 1 and task 2), alongside learning and manipulative capabilities. (2) The process of morphological evolution, highlighting evolutionary models, evolution rates, and morphology at the evolutionary stasis stage. (3) The collective evolutionary trends across multiple morphologies, emphasizing phenomena such as convergent evolution and morphological specialization.

In constructing the hand models, a 50th percentile male hand was used as the base model [52]. The bone segment lengths of this baseline were then replaced with corresponding measurements from hand specimens [56–58], yielding models that preserve the human joint structure but reflect ancient hominin proportions. This substitution procedure follows the same modeling logic and assumptions as in prior studies [40, 55], which similarly integrated bone segment lengths into a unified kinematic model. Given the indeterminacy of joint movements, soft-tissue anatomy, and carpal bones in ancient hominins [40], adopting human joint limits is a common approach [55], to ensure consistency. Keeping joint motion

parameters constant also enables us to focus solely on the contribution of segment length. Finally, to account for natural inter-individual variation, the initial set of models was augmented by injecting Gaussian noise into the specimen-derived segment length measurements. This procedure generated 512 randomized variants for the four specimens, resulting in a total of 4×512 distinct hand morphologies.

Regarding evolutionary parameters, we evaluate 22,528 ($4 \times 512 \times 11$) morphologies in total, with 4×512 morphologies per generation over 11 generations, and each morphology is tested on 64 tasks (32 grasping and 32 manipulation tasks with domain randomization). In each generation, we deterministically retain the top 50% of individuals as elites, while the bottom 50% with lower fitness are discarded. Among the elites, the top 12.5% of the whole population are designated as parents. Each parent then produces four mutated offspring, and these 256 offspring collectively replace the discarded half of the population, keeping the population size constant. In addition to the above evolutionary parameters, the evolutionary process is also constrained by several factors, including the segment lengths in the initial population (Fig. 7a) and the types and numbers of objects specified by the task preferences (Tab. II).

Multiple evolutionary computations can reveal broader phenomena. By applying identical selection pressures to

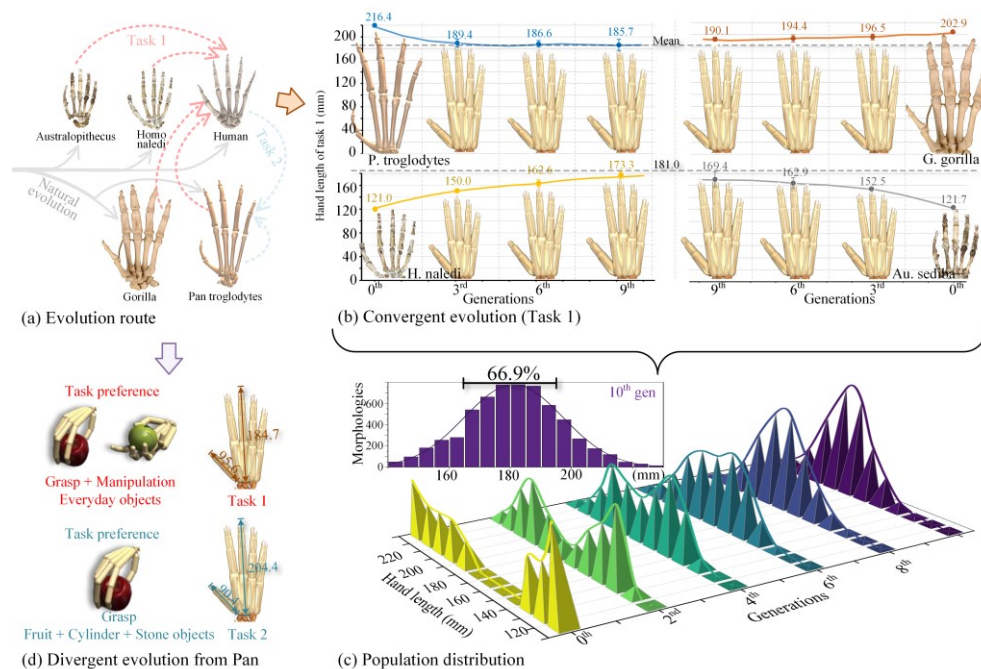


Fig. 7 Morphological evolution of agents with different segment lengths. (a) The phylogenetic relationship of five anthropoid hands and evolutionary pathways validated by the framework. (b) The diverse initial morphologies undergo directional changes and convergent evolution under identical selection pressures. (b) and (c) summarize three independent runs with different random seeds; all numbers denote the mean hand length of the population, and the error bars in (b) indicate ± 1 standard deviation. (c) Convergent change of the population hand-length distribution, pooling morphologies from all three seeds. (d) Hand morphologies diverge from the identical population under different task requirements. In this panel, the red text denotes Task Set 1, whereas the blue text denotes Task Set 2. Task Set 1 comprises both grasping and manipulating a broad range of everyday objects, whereas Task Set 2 comprises grasping-only tasks over fruits, cylinders and stones. Task 2 is included only in this panel to illustrate morphological specialization. Unless otherwise stated, most other experiments in this paper are based on Task Set 1.

closely related species within the same phylogenetic tree, instances of morphological convergent evolution can be observed. Conversely, subjecting a single morphology to different selection pressures can lead to morphological specialization.

Three models have become standard in elucidating the nature of evolutionary divergence: directional change, unbiased random walk, and stasis [59]. Under the selection pressures of Task 1, the diverse initial morphologies undergo directional change towards the inferred morphology (human hand). As the morphologies gradually converge towards the mean value of 181.0 mm, the evolutionary rates decelerate until nearly reaching stasis (Fig. 7b). The variation in population distribution supports this phenomenon (Fig. 7c). From the first to the tenth generation, the proportion of morphologies within the 165 to 195 mm range increases from 11.3% to 66.9%, demonstrating convergent evolution among these closely related species. Furthermore, morphological differentiation is observed under different selection pressures. The object set and task assignments for the two tasks are summarized in Table II. Task 2, which focuses solely on grasping fruits, cylinders and stones, tends to evolve towards ape hand morphologies; whereas Task 1, which involves both grasping and manipulation for a broader range of human daily objects, mostly adapted from the YCB (Yale-CMU-Berkeley) object set [61], tends to evolve towards human hand morphologies (Fig. 7d).

Consequently, the observed directional change in morphology corroborates previous findings that human hands possess a relative advantage in manipulative potential [55]. Furthermore, the morphologies during the stasis phase, being slightly above the inferred morphology, lend support to the

hypothesis of better hand morphology [54].

B. Skill Learning with Different Morphologies

To investigate the role of morphology in learning, and to evaluate whether our results exhibit a Baldwin-like mechanism, we further compared manipulation performance across different generations of evolution. To clearly present the impact of morphology on learning, the controller does not load the model from the last generation, which differs from the approach described in our methodology. Consequently, the controller is reset between every generation, which significantly extends our training time; however, this additional time is well worth it as it allows for a clearer comparison of the effects of morphological differences on both learning speed and ultimate rewards. As morphology evolved, the learning duration decreased, while the ultimate rewards increased (Fig. 8a). This suggests that certain evolved morphologies help alleviate learning impediments, making it easier to explore complex manipulative strategies. Furthermore, the evolution of morphologies and skills contributed to an improvement in grasping and manipulative abilities (Fig. 8b, 8c). These results indicate that morphologies which facilitate learning are implicitly favored by the evolutionary process.

To further quantify how much evolution improves over both the random and biologically inspired baselines, we also analyzed the elite morphologies in each population (Fig. 8(d)). For the random, 0th-generation, and 10th-generation populations, we selected the top-10 morphologies according to their fitness. Compared with the 0th-generation group, the random group can explore a broader range and therefore

TABLE II
OBJECT SET AND TASK ASSIGNMENT

Object name	Shape	Dimensions (mm)	Task usage	Task set in Fig. 7(d)
Block	Box	40×40×40	Grasp & Manipulation	Task Set 1
Stone	Polyhedron	60×60×80	Grasp & Manipulation	Tasks Set 1&2
Tomato Soup Can	Cylinder	70×70×100	Grasp Only	Task Set 1
Tuna Fish Can	Cylinder	85×85×34	Grasp & Manipulation	Task Set 1
Potted Meat Can	Box	100×55×90	Grasp Only	Task Set 1
Strawberry	Sphere	50×50×50	Grasp Only	Task Set 1
Apple	Sphere	80×80×80	Grasp & Manipulation	Tasks Set 1&2
Orange	Sphere	73×73×73	Grasp & Manipulation	Tasks Set 1&2
Peach	Sphere	60×60×60	Grasp Only	Tasks Set 1&2
Mug	Cylinder	90×90×80	Grasp Only	Task Set 1
Cup	Cylinder	57×57×62	Grasp & Manipulation	Task Set 1
Foam Brick	Box	52×52×78	Grasp & Manipulation	Task Set 1
Tennis Ball	Sphere	67×67×67	Grasp & Manipulation	Task Set 1
Golf Ball	Sphere	42×42×42	Grasp Only	Task Set 1
Gelatin Box	Box	70×90×30	Grasp Only	Task Set 1
Baseball	Sphere	73×73×73	Grasp Only	Task Set 1
Cylinder_60 to Cylinder_120	Cylinder	D×D×100, D ∈ {60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 120}	Grasp Only	Task Set 2

The 12 cylinder objects in Task Set 2 (Cylinder_60 to Cylinder_120) were used only for the morphological specialization analysis in Fig. 7(d). The 16 grasping scenarios and 8 manipulation scenarios described elsewhere in this work correspond to the Task Set 1 objects.

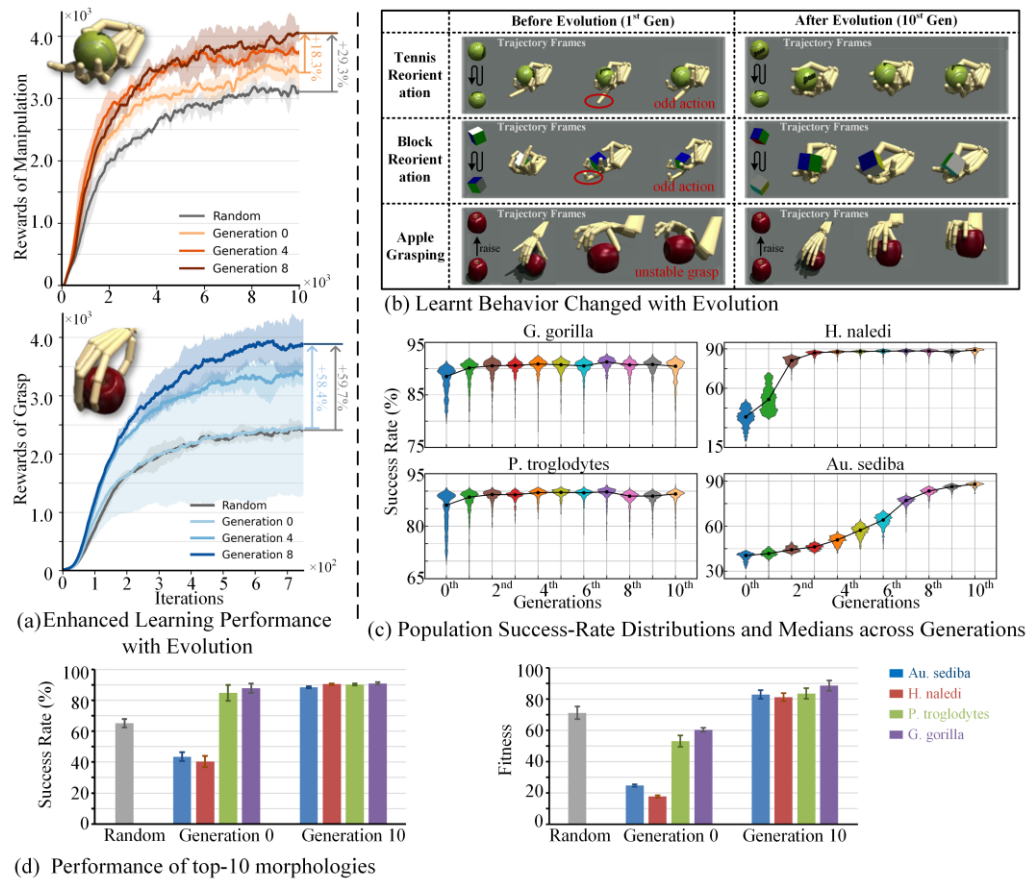


Fig. 8 The impact of morphological evolution on learning and manipulation. (a) Enhanced learning performance with evolution. Learning curves for manipulation (top) and grasping (bottom) under different morphologies. Each trajectory is the mean reward over three independent runs with different random seeds. The grey curve corresponds to the random-morphology baseline, and the colored curves correspond to the populations at Generations 0, 4, and 8. Shaded regions indicate ± 1 standard deviation across runs. (b) Learnt behavior changed with evolution. The evolution of morphologies and skills also affects actual grasping patterns and manipulation gaits. (c) Population success-rate distributions and medians across generations, shown separately for the four species. In each panel, each violin shows the distribution of morphology-level success rates within a generation, and the black line connects the median across generations. The grasping and manipulation success rates are defined in Section II.G. (d) Performance of the selected top-10 morphologies from the random, initial (0th-generation), and final (10th-generation) populations, shown separately for the four species. Bars indicate the mean success rate (left) and fitness (right), and error bars denote one standard deviation.

shows a certain advantage in its top-10 morphologies over the Generation 0 morphologies. However, both groups are inferior to the 10th-generation elites, which exhibit higher success rates and fitness. The fitness metric is defined in Section II.D, whereas the grasping and manipulation success rates provide a more intuitive measure and are defined in Section II.G.

C. Robotic Reconfigurable Hand Evolution

The majority of robotic hands have fundamental differences from human hands. For example, humans possess five fingers, each varying in length and actuation mechanism, whereas robots frequently employ modular fingers, the number and placement of which do not mirror humans. These morphological disparities necessitate additional computation to optimize the performance of hand prototypes.

We extracted segment lengths from human hand specimens

to serve as the initial morphology for the reconfigurable hand. This aims to facilitate the comparison between human-like designs and evolutionary designs in terms of grasping and manipulation performance. In practical applications, any finger length can serve as the initial morphology for the prototype evolution. We use the first 16 objects (Task Set 1 objects) listed in Table II, together with their associated task-usage labels (Grasp Only or Grasp & Manipulation), to define the task set of 16 grasping tasks and 8 manipulation tasks for reconfigurable hand evolution. For efficient parallel execution, the 16 grasping scenarios were duplicated to yield 32 grasping environments, while the 8 manipulation scenarios were repeated four times to yield 32 manipulation environments.

We observed an intriguing phenomenon across all topologies of the hands: the non-thumb fingers became more symmetric in length, with the third and fourth fingers converging to similar lengths, as well as the second and fifth

fingers. This evolutionary phenomenon may be related to the modular design, where each finger possesses identical actuation torque and mechanisms. Previous robotic designs have predominantly focused on replicating human finger lengths, with some even limited to adjusting the mounting positions of modular fingers. However, humans have different muscles for each finger, unlike robots, which use identical modular fingers. Consequently, merely referencing human finger lengths does not necessarily result in the optimal configuration for five-fingered robotic hands. Furthermore, applying human hand morphology to three or four-fingered robotic hands presents additional challenges. Our method offers novel insights into robotic design through the lens of robots' own experience. This optimization process aligns more closely with the intrinsic evolutionary mechanisms of the human hand, rather than merely imitating its external form (Fig. 9).

D. Real-World Evaluation on the Reconfigurable Hand

The evolutionary results further support this view: this unique evolution mechanism enhances the performance of hand prototypes. According to simulation results (Fig. 10a), the average success rate of the optimized designs exceeds that of the baseline human-like designs by approximately 10%. Notably, due to the direct link to the grasping fitness function, the improvements in grasping quality are significantly higher than those in success rate (Fig. 10a). The computed morphological parameters were subsequently transferred to a real-world reconfigurable robot, which demonstrated both the baseline and evolved morphologies, followed by a grasping demonstration (Fig. 10b). To quantitatively assess the hardware benefits of evolution, we equipped the reconfigurable hand with 401-taxel tactile arrays and

dedicated signal-processing PCBs on each finger, and conducted 160 grasping trials on 16 everyday objects, corresponding to those used in simulation (five trials per object for both the baseline and evolved morphologies, i.e., 80 trials per morphology; Fig. 10c). For each trial, a tactile grasp-quality score Q was computed from the pressure distribution; together with the binary outcomes over the five repetitions, this yielded per-object grasp success rates and single-trial grasp-quality scores. The metric Q is consistent with the grasp-quality definition used in Fig. 10a, but here it is evaluated per single trial rather than accumulated over many rollouts due to practical constraints in hardware experiments. Aggregated over all objects, the evolved morphology increased the grasp success rate from 73.75% to 91.25% and raised the mean tactile grasp-quality score from 0.0129 to 0.0213.

E. Generalization to Unseen Objects

To further examine whether the final controller–morphology pairs are genuinely more adaptive, we randomly selected eight everyday objects (Fig. 11) that were not included in the evolutionary–learning cycle, modeled them in SolidWorks as STL meshes, and imported them into the simulation environment. The policies and morphologies were frozen and directly deployed on these unseen objects without any additional training. As summarized in Table III, in simulation the evolved morphology achieved higher mean grasp success rate (69.5% vs. 76.8%), higher mean manipulation success rate (44.0% vs. 48.2%), substantially higher mean grasp quality (206.3 vs. 347.8), and a larger mean number of reorientations per episode (32.6 vs. 48.8) compared with the initial morphology. Note that this result is tested only in simulation, with no real-world transfer.

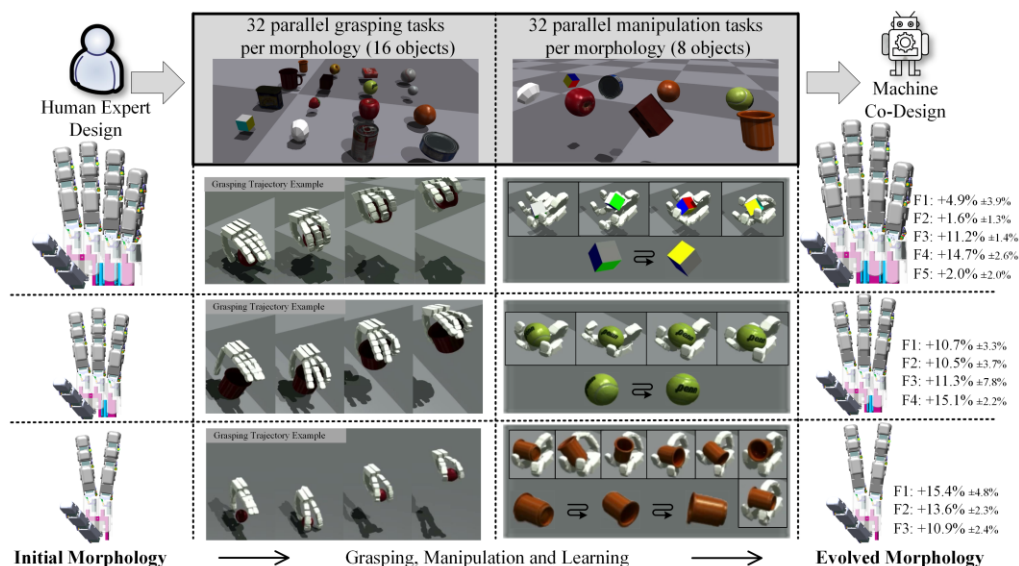


Fig. 9 Reconfigurable hand evolution. It illustrates the evolution of hand prototypes with different numbers of fingers from human-like initial morphologies to new, stable morphologies. The F1–F5 values indicate the relative increase (in %) of each finger's total length in the evolved morphology with respect to the initial morphology, and the accompanying “±” values denote one standard deviation over three runs with different random seeds.

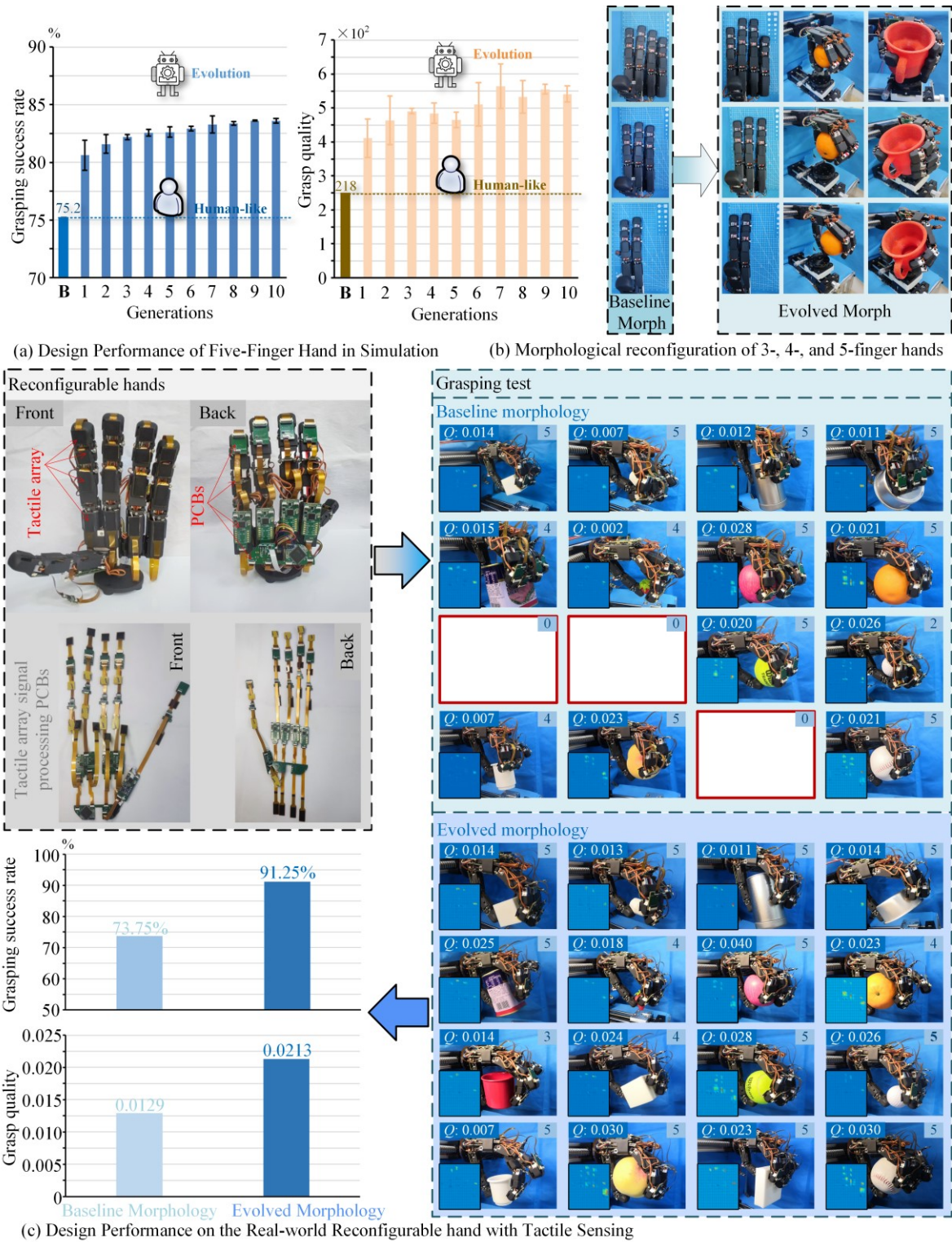


Fig. 10 Design performance comparison of the reconfigurable hand. (a) Simulation results of the five-finger hand, where “B” is the biomimetic (human-like) baseline and the other bars show the mean grasping success rate and grasp quality of the evolved designs, averaged across grasping tasks; error bars indicate one standard deviation over three random seeds. (b) Morphological reconfiguration of the real-world hand from 3-, 4-, to 5-finger layouts, with example grasps for the evolved morphologies. (c) Real-world evaluation with tactile sensing over 160 grasping trials. Left: front and back views of the hand and tactile-array signal-processing PCBs. Right: for baseline and evolved morphologies, each panel shows a representative grasp with its pressure heatmap, the per-object single-trial grasp-quality score Q , and the number of successful trials out of five; the bar plots summarize the average grasping success rate and single-trial grasp quality, where (a) accumulates grasp quality over rollouts to reduce simulation noise.



Fig. 11 Unseen everyday objects used for generalization evaluation. Eight everyday objects are shown, and the numbers indicate the maximum object dimensions (in mm).

TABLE III
PERFORMANCE ON EIGHT UNSEEN OBJECTS

	Initial Morphology	Final Morphology
Mean Grasp Success Rate	69.5%	76.8%
Mean Manipulation Success Rate	44.0%	48.2%
Mean Grasp Quality	206.3	347.8
Mean Number of Reorientations	32.6	48.8

IV. DISCUSSION

In this work, application goals are encoded through the task-preference distribution over grasping and manipulation tasks and over the object set. In the current experiments each of the 16 grasped objects and 8 manipulated objects is assigned equal weight; that is, there is no strong preference for any specific object. This choice encourages the evolved hand morphology to remain broadly adaptive across the objects. For moderate changes within a similar family of tasks (e.g., moderately adjusting the weighting of existing tasks), the evolved hand morphology can be kept fixed, and the shared controller can be fine-tuned under the new task-preference distribution. For more substantial changes in tasks (e.g., reducing the weighting of some existing tasks to zero), the evolutionary outer loop can be warm started from the previously evolved morphologies together with the pre-trained controller, which reduces the computational cost of adaptation compared to restarting from a random population. Finally, if the task set changes drastically (e.g., the introduction of entirely new task families), the originally evolved morphology may no longer be suitable, and re-running the full co-optimization from scratch is likely to be more appropriate.

Moreover, Fig. 8 suggests that morphological changes are associated with learning speed. Over the course of human hand evolution, increasingly sophisticated tool-use skills have been accompanied by changes in specific mechanical properties, such as thumb opposability and finger-length ratios. Across the four fossil-inspired populations in Fig. 7, we observe a systematic increase in relative thumb length, defined as the total length of the thumb metacarpal and phalanges divided by that of the middle finger. This ratio increases from 0.614 to 0.667 for *Au. sediba*, from 0.402 to 0.512 for *P. troglodytes*, from 0.620 to 0.669 for *H. naledi*, and from 0.503 to 0.533 for *G. gorilla*. Because the initial morphology of the reconfigurable hand is based on the human hand, which

already has a relatively long thumb (ratio 0.656), the thumb-to-other-finger length ratio of the final morphologies varies only within ± 0.02 across the three-, four-, and five-finger hands. A relatively longer and more opposable thumb enlarges the overlap between the thumb and finger workspaces and mechanically stabilizes opposition grasps, thereby making successful grasping and manipulation policies easier to discover. This, in turn, creates a form of selection pressure: hand morphologies with such mechanical properties are preferentially retained because they better support stable grasps and complex tool use, giving rise to a Baldwin-like mechanism in our robotic and simulation settings.

However, our framework also differs in several important ways from the canonical Baldwin effect observed in nature. First, all candidate morphologies share a common Transformer-based controller, so experience collected across different hand morphologies is used to update a single shared “brain”, whereas biological organisms do not share a global brain and learning is strictly individual. Second, in our framework evolution acts only on morphology parameters, while learning is implemented as gradient-based updates to the shared controller under a fixed training schedule; selection therefore favors morphologies that make learning more effective within this budget, rather than directly optimizing heritable learning abilities as in biological evolution. For these reasons, we describe our setting as exhibiting a Baldwin-like or Baldwin-effect-inspired mechanism, rather than claiming a direct realization of the standard Baldwin effect in nature.

In terms of prototype evolution, robots have demonstrated unique evolutionary pathways and exceeded biomimetic baselines. With autonomously designed reward functions [61] using large models, the advancement of 3D printing technology, and the optimization methods for general morphological parameters, the design abilities of future robots could be significantly enhanced, possibly no longer requiring human assistance. This capability could become a critical feature of autonomous robots in the future, enabling them to become aware of their own physicality and to compute and adjust critical bodily parameters. This prospect is intriguing when considering the millions of years of human evolutionary history. One might speculate whether, millions of years from now, robots will also possess long-term evolutionary abilities to respond to environmental changes and what their final physical morphologies might be.

The present work is limited to fully actuated hands. In future research, the two-level nested co-optimization framework could in principle be reused for underactuated hands, but the morphology parameterization would need to be enriched with underactuation-related variables (e.g., link and tendon parameters). For underactuated hands, under actuation restricts the set of achievable contact forces to a low-dimensional subset of the wrench space implicitly assumed in this metric. A potential extension would be to incorporate the actuation mapping from motor inputs to joint torques and define grasp-quality measures on the resulting feasible wrench space. In addition, the inner-loop observation space, the

information encoded at each node of the hand graph, and the action space would also have to be modified, since actuators no longer provide independent control at each joint. Searching over the additional underactuation-related design parameters would further increase the computational cost of the outer-loop evolution.

Another limitation of this work lies in the hardware constraints. Except for the thumb, all fingers are built from similar modular units with fixed mass, actuation properties, and joint limits, and evolution is restricted to a design space spanned only by link lengths and mounting positions. This fixed topology (e.g., the number and orientations of joints) constrains the directions of finger motion and thus the overlap of workspaces. Moreover, because the inertia and torque capabilities are similar across fingers—unlike in biological hands—the evolved morphology of a modular-finger hand differs from that of the human hand. This raises a broader question: given that most robotic hands are constructed from modular fingers, is a human-hand-like layout necessary? In addition, under a fixed five-finger topology, our current evolution framework can also be viewed as parameter optimization for the robot hand.

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